Chapter 05

User & Service Management

Episode 5.01

Episode title: **Managing Files & Directories**

Objective: 2.1

Files and Directories in Linux

Filesystem hierarchy

Files vs directories

Inodes and metadata

File Management Commands

`ls` and `stat`

`cp` and `mv`

`rm` (safe usage)

File & Directory Utilities – Exam Coverage

Navigation:

- cd pwd ls

File creation & metadata:

- touch stat file

Directory management:

- mkdir rmdir

File comparison:

- diff sdiff

Searching for Files and Open Resources

`find` – Search filesystem in real time

`locate` – Search indexed file database

`lsof` – List open files and processes

Common troubleshooting scenarios

Links in Linux

Hard links

Symbolic links

Use cases

Managing Files & Directories (DEMO)

Create, copy, move, remove files

Create hard and symbolic links

Verify results

Episode 5.02

Episode title: **GUI File Management**

Objective: 2.1

Common Linux GUI File Managers

GNOME Files (Nautilus)

KDE Dolphin

Thunar

Caja

GUI File Operations and Safety

Copy and move operations

Delete behavior

Permissions and ownership

GUI vs Command Line File Management

Ease of use

Speed and automation

Remote administration

GUI File Management (DEMO)

Browse filesystem

Create, copy, move, delete files

Inspect permissions

Episode 5.03

Episode title: **Local User Accounts**

Objective: 2.2

Users, Groups, and IDs

Users and groups

UID and GID

Primary vs supplementary groups

User and Group Management Commands

useradd | usermod | userdel

groupadd | groupmod | groupdel

User Information and Session Commands

Identity and groups: - id groups whoami

Login and session tracking: - who w last lastlog

Directory-based lookup: - getent passwd

Account Management Commands – Exam

Add users and groups: - adduser useradd groupadd

Modify accounts: - usermod groupmod chsh passwd

Remove accounts: - deluser userdel groupdel

Account Security and Best Practices

Least privilege

Account lifecycle

System vs regular users

Managing Local User Accounts (DEMO)

Create, modify, delete users

Create and manage groups

Verify results

Episode 5.04

Episode title: **Account Files & Attributes**

Objective: 2.2

Account Configuration Files

/etc/passwd

/etc/shadow

/etc/group

User Profile Templates

`/etc/skel` – New user home templates

`/etc/profile` – System-wide environment settings

Applied during user creation

/etc/passwd Explained

/etc/passwd structure

Fields and meanings

Login shells

/etc/shadow Explained

/etc/shadow purpose

Password hashes

Account aging

Account Locking, Aging, and Expiration

Locking accounts:

- passwd –l
- usermod –L

Password aging:

- chage

Expiration settings:

- Account
- password

UID, GID, and System Accounts

UID and GID ranges

System vs regular users

Service accounts

Inspecting Account Files & Attributes (DEMO)

View passwd, shadow, group files

Verify UID/GID

Identify account types

Episode 5.06

Episode title: **Processes & Jobs**

Objective: 2.3

Intro to Linux Processes & Jobs

Running program – process

Multiuser, multitasking

Jobs

Job control

Processes in Linux

Process IDs (PID)

Parent/child processes

Process states

Process Limits and Resource Constraints

Limits on CPU, memory, and processes

Prevents resource exhaustion

Configured per user. or system-wide

Monitoring and Controlling Processes

ps and top

Signals and kill

nice and renice

Process Inspection Tools – Exam Coverage

Process viewers: top htop atop pstree

Detailed inspection: - ps /proc/<PID> lsof strace

Performance tools: - mpstat pidstat

Jobs and Scheduling

Foreground/background jobs

jobs fg bg

Basic scheduling concepts

Linux Scheduling Tools – Exam Coverage

- cron/crontab – Recurring jobs
- at – One-time jobs
- anacron – Jobs on intermittently powered systems

Managing Processes & Jobs (DEMO)

Monitor processes

Control jobs

Send signals

Adjust priorities

Episode 5.07

Episode title: **Package Management**

Objective: 2.4

Package Management Concepts

Packages and dependencies

Repositories

Metadata and signatures

Repository Management & Trust

- Enable or disable repositories - Third-party repositories
- GPG signatures and trust
- Package and repository exclusions

Managing Software Alternatives

- Multiple versions of same software
- System-wide default selection
- Common with editors and runtimes

Common Linux Package Managers

- apt (Debian/Ubuntu)
- dnf/yum (RHEL/Fedora)
- zypper (SUSE)

Language-Specifc Package Managers

- `pip` – Python packages
- `npm` – JavaScript packages
- `cargo` – Rust packages
- Operate outside system package managers

Package Lifecycle and Safety

Install, update, remove

System updates

Rollback considerations

Sandboxed Applications

- Application isolation
- Reduced system impact
- Examples: Snap, Flatpak
- Trade-offs vs traditional packages

Package Management (DEMO)

Search, install, update, remove packages

Verify package state

Episode 5.08

Episode title: **Service Configuration**

Objective: 2.4

Service Config Intro

Configuring network services

More than editing files

Interaction awareness

Common Linux Network Services

DNS

DHCP

HTTP

SMTP

Common Service Implementations - Exam

- DNS: bind, systemd-resolved
- Time sync: NTP / PTP
- HTTP: Apache (httpd), Nginx
- Mail access: IMAP4

Service Configuation and Management

Configuration files

systemd service units

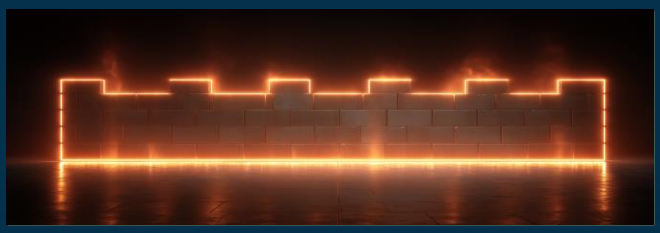
Enable vs start

Service Security and Verification

Least privilege

Firewalls

Logs and testing



Service Configuration Workflow (DEMO)

Install

Configure

Start

Verify